Reflective report: ethical, social and professional impacts of cloud computing

Before starting this module, I mostly thought of cloud computing as a technical choice about where systems run. Working through the units changed that. Each practical task, from writing Azure provisioning scripts to running a security audit on a Docker container, showed me that the decisions behind cloud systems also affect people, organisations and society, and that these effects are not always obvious at the start.

The ethical issue that stood out most for me was data protection. When I planned the cloud strategy for my hypothetical reinsurer, I realised that choosing a storage location or encryption method is also a decision about whose data is at risk if something goes wrong. Under UK GDPR, organisations remain responsible for personal data even when a cloud provider stores it (Information Commissioner's Office, 2023). The shared responsibility model made this clearer for me, because the provider secures the infrastructure but the customer is still accountable for configuration and access. My Docker audit backed this up. Most of the issues I found came from default settings that nobody had changed, which taught me that ethical responsibility often comes down to careful, everyday work rather than big decisions.

On the social side, I was surprised by how much power is concentrated in a small number of providers. Carugati (2023) argues that the largest cloud companies are also becoming the main suppliers of generative AI, which could strengthen their position further. This matters socially because smaller organisations may have fewer real choices, and because outages or policy changes at one provider can affect millions of people at once. The environmental impact is another social concern. The International Energy Agency (2024) reports that data centres already use a significant share of global electricity, and this is expected to grow with AI. I had not considered this before, and it has made me think more about switching off unused resources, which I had treated mainly as a cost saving.

Professionally, the module showed me how important standards and documentation are. Mapping my audit findings to ISO/IEC 27001 controls helped me explain why each issue mattered, rather than just listing problems. The BCS Code of Conduct also reminds IT professionals to act in the public interest and only take on work within their competence (BCS, 2022). I found this relevant when writing code for the e-learning task, because I had to be honest about what my implementation did and did not secure.

These factors clearly influence real world decisions. Sharma and Kumar (2024) describe hybrid cloud as a way for organisations to balance flexibility with control, and in my own report I chose hybrid partly because of regulatory and data concerns rather than technical ones alone. Alshareef (2023) also lists security, privacy and cost as ongoing challenges, which matches what I found in practice.

Looking ahead, I want to treat ethical and social questions as part of design from the beginning, not something to review at the end. In future projects I plan to include data protection, environmental impact and provider dependency in my early planning alongside cost and performance.

References

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